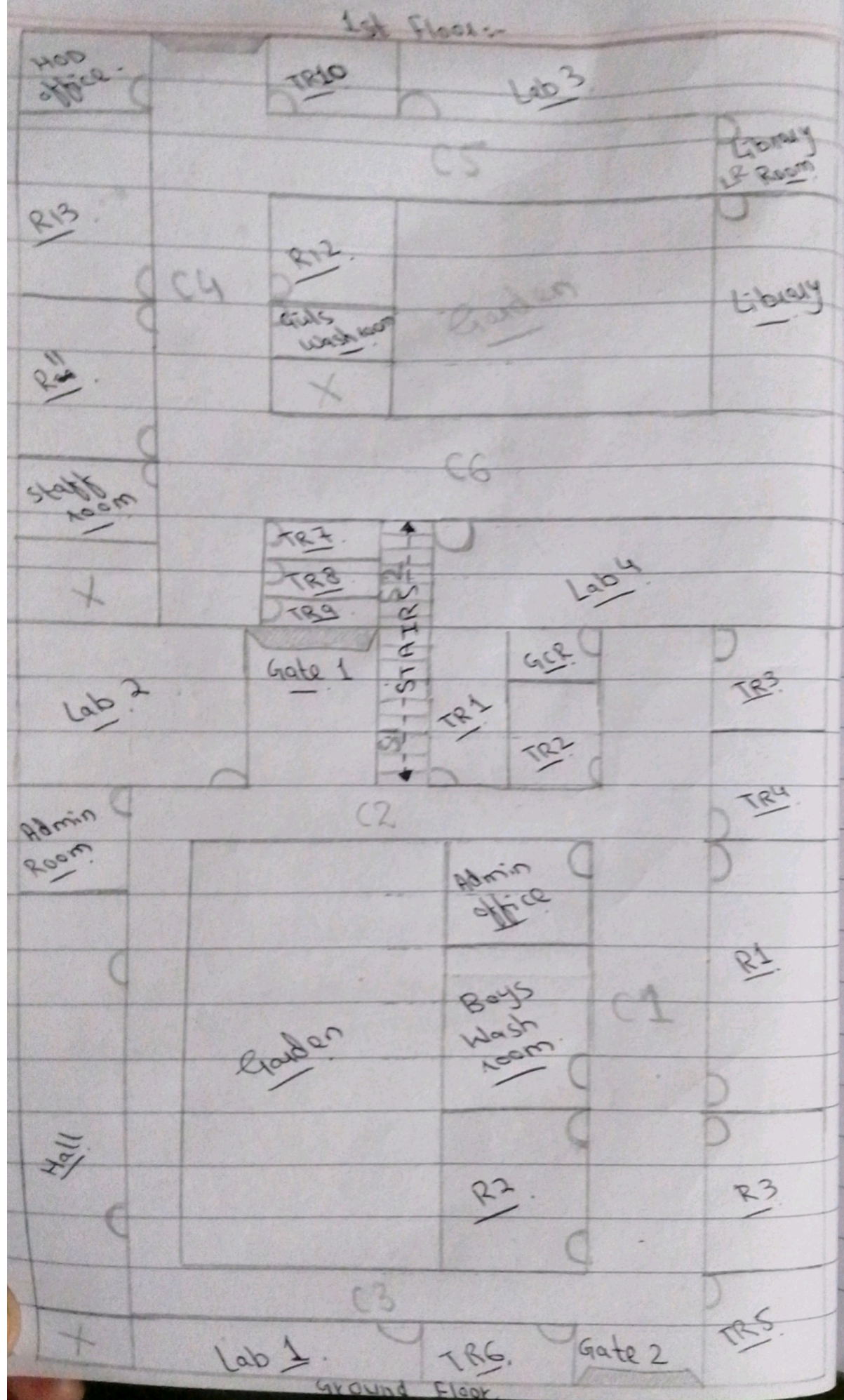
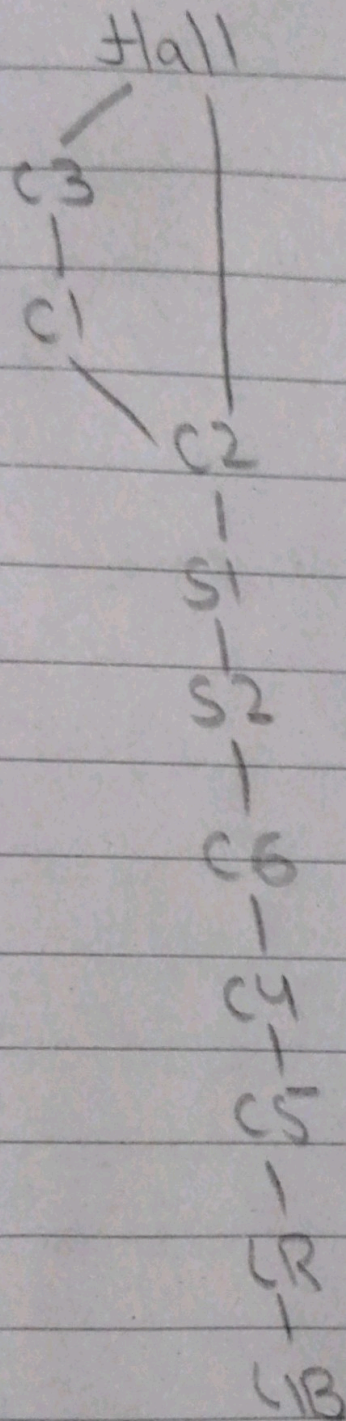


* LAB ACTIVITY : 05 :- - Map of CS Department :-



• Hall to Library:-



-Transition Table:-

Step:	From:	To:
1	Hall	C3, C2
2	C3	C1
3	C1	C2
4	C2	S1
5	S1	S2
6	S2	C6
7	C6	C4
8	C4	C5
9	C5	LR
10	LR	LIB

LR (Library Room)
S1, S2 (stairs).

Q: Find path Hall to Library:-

Q: [Hall]

V: [Hall]

→ Dequeue Hall & expand:

Q: [C3, C2]

V: [Hall, C3, C2]

→ Dequeue C3 & expand:

Q: [C1]

V: [Hall, C3, C2, C1]

→ Dequeue C1 & expand:

Q: [C2]

V: [Hall, C3, C2, C1]

→ Dequeue C2 & expand:

Q: [S1]

V: [Hall, C3, C2, C1, S1]

→ Dequeue S1 & expand:

Q: [S2]

V: [Hall, C3, C2, C1, S1, S2]

→ Dequeue S2 & expand:

Q: [C6]

V: [Hall, C3, C2, C1, S1, S2, C6]

→ Dequeue C6 & expand:

Q: [C4]

V: [Hall, C3, C2, C1, S1, S2, C6, C4]

→ Dequeue c_4 & expand:

$Q: [c_5]$

$V: [Hall, c_3, c_2, c_1, s_1, s_2, c_6, c_4, c_5]$

→ Dequeue c_5 & expand:

$Q: [LR]$

$V: [Hall, c_3, c_2, c_1, s_1, s_2, c_6, c_4, c_5, LR]$

→ Dequeue LR & expand:

$Q: [LIB]$

$V: [Hall, c_3, c_2, c_1, s_1, s_2, c_6, c_4, c_5, LR, LIB]$

* FORMULATION OF PROBLEM:-

- Initial State: Hall.
- Goal State: Library.
- State space: CS Department.
- Action: Move from Hall to Library.
- Path cost: 1 for each move.